Nuclear Medicine and Radiation Safety

Answer key with worked solutions

A. Understanding

1. Units.

> (a) **Becquerel:** 1 decay per second (activity). **Gray:** 1 J of energy absorbed per kg of tissue (absorbed dose). **Sievert:** absorbed dose × radiation weighting factor (equivalent dose, reflecting harm). (b) Alpha particles deposit their energy along a **very short, dense track**, causing many ionizations close together, which damages DNA far more severely than the same energy spread out by gamma rays.

2. Choosing an isotope.

> **Technetium-99m**: it emits **gamma rays** that leave the body to reach the camera, and its **short half-life** (6.01 h) keeps the dose low. Iodine-131's beta particles add dose without helping the image, and its half-life is longer. Radium-223 emits **alpha** particles, which cannot be detected outside the body and would damage tissue. Carbon-14 emits only beta particles and its half-life (5730 yr) is far too long.

B. Dose

3. Tissue dose.

> (a) $\dfrac{1.20 \times 10^{-3}\ \text{J}}{0.0250\ \text{kg}} = 0.0480\ \text{J/kg} = \mathbf{0.0480\ Gy}$
>
> (b) Beta (factor 1): $0.0480\ \text{Gy} \times 1 = 0.0480\ \text{Sv} = \mathbf{48.0\ mSv}$
>
> (c) Alpha (factor 20): $0.0480\ \text{Gy} \times 20 = 0.960\ \text{Sv} = \mathbf{960.\ mSv}$

C. Protection

4. Inverse-square law.

> (a) $40.0\ \mu\text{Sv/h} \times \left(\dfrac{1.00\ \text{m}}{0.500\ \text{m}}\right)^2 = 40.0\ \mu\text{Sv/h} \times 4.00 = \mathbf{160.\ \mu Sv/h}$
>
> (b) $\left(\dfrac{1.00\ \text{m}}{d}\right)^2 = \dfrac{1.00\ \mu\text{Sv/h}}{40.0\ \mu\text{Sv/h}}$, so $d = 1.00\ \text{m} \times \sqrt{40.0} = \mathbf{6.32\ m}$

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5. Dose from time.

> $t = 15.0\ \text{min} \times \dfrac{1\ \text{h}}{60\ \text{min}} = 0.250\ \text{h}$; dose $= 10.0\ \mu\text{Sv/h} \times 0.250\ \text{h} =$ **2.50 µSv**

6. Shielding.

> (a) $\dfrac{1.20\ \text{mm}}{0.30\ \text{mm}} = 4$ half-value thicknesses; $\left(\frac{1}{2}\right)^4 = 0.0625$, so **6.25 %** passes through
> (b) $\frac{1}{32} = \left(\frac{1}{2}\right)^5$: 5 half-value thicknesses $= 5 \times 0.30\ \text{mm} =$ **1.5 mm**

D. Isotopes in use

7. Fluorine-18.

> $t = 2.00\ \text{h} = 120.\ \text{min}$; $n = \dfrac{120.\ \text{min}}{110\ \text{min}} = 1.091$
> $A = 370\ \text{MBq} \times \left(\frac{1}{2}\right)^{1.091} =$ **174 MBq**

8. Annihilation energy.

> $E = mc^2 = (9.109 \times 10^{-31}\ \text{kg})(2.998 \times 10^{8}\ \text{m/s})^2 = 8.187 \times 10^{-14}\ \text{J}$
> $\dfrac{8.187 \times 10^{-14}\ \text{J}}{1.602 \times 10^{-13}\ \text{J/MeV}} = 0.511\ \text{MeV} =$ **511 keV** ✓ (a positron has the same mass, so each of the two gamma rays carries 511 keV)

E. Challenge

9. Benefit and risk.

> (a) $\dfrac{7.0\ \text{mSv}}{2.4\ \text{mSv/yr}} =$ **2.9 years** of natural background. (b) The extra risk from this dose is small, while the scan can **diagnose a serious illness** early and guide treatment; when the expected **benefit outweighs the risk**, the scan is justified. Doctors still keep the dose as low as reasonably achievable and avoid unnecessary scans.

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